

Real-World Outcomes of Transcatheter Tricuspid Valve Replacement Analysis From the STS/ACC TVT Registry

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 [Supplemental content](#)

IMPORTANCE Transcatheter tricuspid valve replacement (TTVR) demonstrated superior outcomes over medical therapy in patients with severe tricuspid regurgitation (TR) in the Edwards EVOQUE Transcatheter Tricuspid Valve Replacement: Pivotal Clinical Investigation of Safety and Clinical Efficacy Using a Novel Device II (TRISCEND II) randomized clinical trial, and received regulatory approval in the US in 2024. Contemporary real-world data on its effectiveness and safety remain limited.

OBJECTIVE To evaluate 30-day clinical, echocardiographic, and health status outcomes of TTVR in real-world use.

DESIGN, SETTING, AND POPULATION Retrospective cohort study of all consecutive patients who underwent TTVR in the US from February 2024 through March 2025 in the Society of Thoracic Surgeons/American College of Cardiology Transcatheter Valve Therapy Registry. Patients had symptomatic, severe TR despite optimal medical therapy and TTVR was deemed appropriate by a heart team. Statistical analysis was conducted from September 2025 to February 2026.

EXPOSURE Device-enabled TTVR.

MAIN OUTCOMES AND MEASURES Thirty-day event rates (all-cause death, stroke, bleeding, new cardiac implantable electronic device [CIED] implantation, heart failure hospitalizations), TR reduction, and changes in health status (New York Heart Association [NYHA] functional class and Kansas City Cardiomyopathy Questionnaire Overall Summary [KCCQ-OS] score) are reported. Subgroup analyses examined the impact of baseline CIED status on outcomes.

RESULTS Among 1034 attempted procedures at 82 centers (mean [SD] age, 77.1 [10.6] years; 69.1% female; 73.2% NYHA functional class III/IV), a valve was successfully implanted in 1017 patients (98.4%). Mild or less TR was achieved in 98.4% of patients post procedure and in 97.7% at 30 days. At 30 days, all-cause mortality was 3.1%; stroke, 0.2%; bleeding, 7.9%; new CIED, 15.9% in CIED-naïve patients; and heart failure hospitalization, 3.1%. There were significant improvements in NYHA functional class (class I/II, 82.7%; $P < .001$) and mean KCCQ-OS score (22.4 points; $P < .001$) from baseline to 30 days. There were no significant differences in 30-day mortality ($P = .47$), heart failure hospitalization ($P > .99$), and functional outcomes ($P = .55$) when patients were stratified by baseline CIED status.

CONCLUSIONS AND RELEVANCE Early US real-world experience with TTVR confirms safety and effectiveness in patients with severe TR. Thirty-day outcomes are consistent with the TRISCEND II pivotal trial, demonstrating acceptable safety, near-complete TR elimination, and significant health status improvements in an older, comorbid population. Rates of new CIED implantation and bleeding were lower than randomized clinical trial experience.

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Tricuspid regurgitation (TR) is the most prevalent valvular heart disease in older adults, with moderate or greater TR found in 3.7% of US adults aged 65 to 85 years,¹ often leading to worsening right heart failure, recurrent heart failure hospitalization, and an increased risk of mortality.²⁻⁴ Surgical repair or replacement remains uncommon, primarily due to a high burden of comorbidities and operative risks.⁵⁻⁷ Although tricuspid transcatheter edge-to-edge repair (T-TEER) can improve symptoms, this procedure rarely eliminates TR and worsening or persistent TR after the procedure is associated with worse outcomes.⁸⁻¹¹

Transcatheter tricuspid valve replacement (TTVR) has emerged as a novel therapeutic option to treat TR. The EVOQUE TTVR system (Edwards Lifesciences), evaluated in the Edwards EVOQUE Transcatheter Tricuspid Valve Replacement: Pivotal Clinical Investigation of Safety and Clinical Efficacy Using a Novel Device II (TRISCEND II) randomized clinical trial, demonstrated near elimination of TR and superiority in the composite safety and effectiveness end point over medical therapy alone at 1 year, driven primarily by improvements in symptoms and quality of life.¹² However, more than 10% of patients experienced major bleeding and more than 15% required cardiac implantable electronic devices (CIEDs), raising questions about how these trade-offs will translate to routine clinical practice. Furthermore, TTVR may be more challenging in patients with a preexisting CIED because it can impair visualization and complicate valve deployment, and data regarding procedural performance and outcomes in patients with vs without baseline CIEDs are limited.¹³

The TTVR system was approved by the US Food and Drug Administration on February 1, 2024. This study reports the first US real-world TTVR experience from the national Society of Thoracic Surgeons/American College of Cardiology Transcatheter Valve Therapy (STS/ACC TVT) Registry to evaluate safety and effectiveness outcomes at 30 days and outcomes stratified by the presence of CIED at baseline.

Methods

Design, Setting, and Study Population

This is a retrospective cohort study of all consecutive patients who received TTVR in the US from February 2024 through March 2025 in the STS/ACC TVT Registry. Patients had symptomatic, severe TR despite optimal medical therapy and TTVR was deemed appropriate by a heart team. There were no inclusion or exclusion criteria for the study. Participation in the registry is required for Medicare coverage under the TTVR National Coverage Determination, with a Coverage With Evidence Development requirement. The registry captures patient demographics, history, risk factors, cardiac and health status, indications for the procedure, procedural measures, and adverse event rates. Clinical status and adverse events are also recorded during the index hospitalization and 30 days after the index procedure, using standardized STS/ACC TVT Registry data collection forms and definitions.¹⁴ Data were analyzed for all patients with an attempted TTVR procedure, defined by the

Key Points

Question In early US real-world registry experience, what are the effectiveness, safety, and patient-reported outcomes of transcatheter tricuspid valve replacement (TTVR)?

Findings Among 1034 patients undergoing TTVR across 82 centers, a valve was successfully implanted in 98.4%, and residual mild or less tricuspid regurgitation was achieved in 97.7% of patients at 30 days. At 30 days, all-cause death was 3.1%; stroke, 0.2%; bleeding, 7.9%; and need for a new cardiovascular implantable electronic device among patients without a preexisting device was 15.9%.

Meaning In the first year after approval in the US, patients undergoing TTVR experienced acceptable early safety, near elimination of tricuspid regurgitation, and significant symptomatic and health status improvements.

initiation of vascular access or skin puncture. The reporting of this study was informed by the principles outlined in the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) guidelines.

Study Oversight

The STS/ACC TVT Registry has approval from a central institutional review board (Advarra) and the waiver for informed consent was granted under the Common Rule (45 CFR §46). Individual sites are responsible for data entry, and the STS/ACC TVT Registry coordinates data collection and quality oversight.

Device and Procedure

The device is the first and only approved TTVR system worldwide, with an indication for patients with symptomatic, severe TR despite optimal medical therapy for whom TTVR is deemed appropriate by a heart team (eFigure 1 in [Supplement 1](#)). The trileaflet bovine pericardial valve is mounted in a self-expanding nitinol frame with an intra-annular fabric sealing skirt and 9 anchors that engage the leaflets, annulus, and subvalvular apparatus for fixation. The valve is delivered via transfemoral venous access through a 28F transcatheter delivery system with 3 planes of flexion for controlled maneuverability. Four valve sizes (44 mm, 48 mm, 52 mm, 56 mm) are available, with size selected primarily by computed tomography-derived annular measurements. Procedures are typically performed under general anesthesia with multimodality imaging (ie, transesophageal echocardiography and/or intracardiac echocardiography with fluoroscopy).¹²

Covariates

Baseline demographic, clinical, procedural, and echocardiographic variables were collected using standardized forms. TR etiology and severity were site assessed per registry definitions. Surgical risk was classified by the heart team (extreme, high, intermediate, or low risk). Baseline health status was assessed via the Kansas City Cardiomyopathy Questionnaire Overall Summary (KCCQ-OS) score. Procedural and clinical variables were captured at discharge and 30 days.

Table 1. Baseline Patient Characteristics

Characteristic	No. (%) [N = 1034]
Demographics	
Age, mean (SD), y	77.1 (10.6)
Age categories, y	
<65	104 (10.1)
65-74	209 (20.2)
75-84	492 (47.6)
≥85	229 (22.1)
Sex	
Male	320 (30.9)
Female	714 (69.1)
Race and ethnicity, No./total No. (%)	
American Indian or Alaska Native	4/959 (0.4)
Asian	50/959 (5.2)
Black or African American	104/959 (10.8)
Hispanic or Latino	67/1013 (6.6)
White	798/959 (83.2)
Multiracial	3/959 (0.3)
BMI, mean (SD)	27.1 (12.2)
NYHA functional class, No./total No. (%)	
I (No symptoms with normal activity, healthiest)	14/1020 (1.4)
II (Symptoms with normal activity)	259/1020 (25.4)
III (Symptoms with minimal activity)	662/1020 (64.9)
IV (Symptoms even at rest, sickest)	85/1020 (8.3)
Medical history/comorbidities	
Heart failure	941 (91.0)
Hypertension	881 (85.2)
Atrial fibrillation/atrial flutter	867 (83.8)
Heart failure hospitalization within past year, No./total No. (%)	355/734 (48.4)
CIED (including pacemaker)	393 (38.0)
QRS duration >120 ms, No./total No. (%)	239/721 (33.1)
Diabetes	307 (29.7)
Cardiomyopathy	235 (22.7)
Chronic lung disease	233 (22.5)
Stroke	141 (13.6)
Myocardial infarction	137 (13.2)
Liver disease	127 (12.3)
Peripheral artery disease	92 (8.9)
Transient ischemic attack, No./total No. (%)	71/1026 (6.9)
Currently receiving dialysis, No./total No. (%)	64/1033 (6.2)
Carotid artery stenosis	44 (4.3)
Endocarditis	27 (2.6)
Medications, No./total No. (%)	
Diuretic	865/1031 (83.9)
Anticoagulant	702/1031 (68.1)
Antiplatelet	372/1031 (36.1)
Laboratory findings	
Hemoglobin, mean (SD), g/dL	12.2 (2.0)
Creatinine, mean (SD), mg/dL	1.5 (1.3)
Creatinine ≥1.5 mg/dL	294 (28.8)

(continued)

Table 1. Baseline Patient Characteristics (continued)

Characteristic	No. (%) [N = 1034]
Echocardiography measures	
Tricuspid valve disease etiology, No./total No. (%) ^a	
Primary	412/999 (41.2)
Secondary	432/999 (43.2)
Pacemaker induced	94/999 (9.4)
Other	61/999 (6.1)
Severe or greater tricuspid regurgitation, No./total No. (%)	996/1030 (96.7)
Tricuspid valve annulus size, mean (SD), mm	41.8 (14.7)
End-diastolic mid-RV diameter, mean (SD), cm	3.9 (1.0)
End-diastolic basal RV diameter, mean (SD), cm	4.8 (0.9)
RV systolic pressure, mean (SD), mm Hg	41.7 (14.0)
LVEF, mean (SD), %	55.9 (9.8)
Moderate/severe mitral regurgitation, No./total No. (%)	356/892 (39.9)

Abbreviations: BMI, body mass index (calculated as weight in kilograms divided by height in meters squared); CIED, cardiac implantable electronic device; LVEF, left ventricular ejection fraction; NYHA, New York Heart Association; RV, right ventricle.

SI conversion factors: to convert hemoglobin to g/L, multiply by 10.0; creatinine to μmol/L, multiply by 88.4.

^a Primary tricuspid valve disease was defined as regurgitation caused by structural valve abnormalities, whereas secondary tricuspid valve disease was defined as regurgitation in the setting of a structurally normal valve with other conditions (eg, myocardial infarction or cardiomyopathy) compromising the valve's ability to function normally. Any remaining etiologies were categorized based on the site's discretion.

Outcomes

Safety outcomes assessed through 30 days were all-cause and cardiovascular mortality, stroke, bleeding, new CIED implantation (permanent pacemaker or implantable cardioverter-defibrillator), tricuspid valve reintervention, major vascular complications, device migration or embolization, and heart failure hospitalization. Bleeding comprised access site, gastrointestinal, genitourinary, hematoma at access site, retroperitoneal, hemorrhagic stroke, and other. The effectiveness outcome was residual TR assessed by transthoracic echocardiography at post procedure or discharge and 30 days. Functional and quality-of-life measures were assessed via New York Heart Association (NYHA) functional class and KCCQ-OS score. Because interactions with preexisting leads may influence both procedural strategy and outcomes, the cohort was stratified by CIED presence at baseline.

Statistical Analysis

Statistical analysis was conducted from September 2025 to February 2026. Summary statistics are presented as means (SDs) or medians (IQRs) for continuous variables and as counts with percentages for categorical variables. Within-patient changes were assessed using paired *t* tests and Wilcoxon signed-rank tests. All descriptive analyses were performed on a complete-case basis. To evaluate robustness to missing data, inverse probability weighting sensitivity analyses were performed for TR severity, NYHA functional class, and KCCQ-OS score. Methodological details are provided in the eMethods in

Table 2. Procedural Characteristics^a

Characteristic	No. (%) [N = 1034]
Procedure indication	
Tricuspid regurgitation	1030 (99.6)
Tricuspid stenosis	4 (0.4)
Procedure duration, median (IQR), min	103.0 (81.0-130.0)
Type of anesthesia	
General anesthesia	1019 (98.5)
Deep sedation/analgesia	2 (0.2)
Moderate sedation/analgesia	13 (1.3)
Access site	
Femoral vein	1007 (97.4)
Jugular vein	26 (2.5)
Device size, No./total No. (%), mm	
44	202/1024 (19.7)
48	307/1024 (30.0)
52	357/1024 (34.9)
56	158/1024 (15.4)
Device implanted successfully ^b	1017 (98.4)
Device deployment attempts, No.	
1	1024 (99.0)
2	9 (0.9)
3	1 (0.1)
Conversion to surgery, No./total No. (%) ^c	8/1032 (0.8)
Concomitant procedure, No./total No. (%) ^d	48/1032 (4.7)
Mechanical support, No./total No. (%) ^e	8/1031 (0.8)
Hospital length of stay, median (IQR), d	
Procedure to discharge	2.0 (2.0-5.0)
Admission to discharge	4.0 (2.0-6.0)
Discharge location, No./total No. (%)	
Home	950/1013 (93.8)
Other acute care hospital	3/1013 (0.3)
Skilled nursing facility	26/1013 (2.6)
Left against medical advice	2/1013 (0.2)
Extended care/transitional care unit/rehabilitation	23/1013 (2.3)
Other discharge location	9/1013 (0.9)

^a A procedure was considered to have been attempted in all patients entered into the Society of Thoracic Surgeons/American College of Cardiology Transcatheter Valve Therapy Registry. Patients were considered to have a valve implanted if the procedure was initiated and not aborted and there was no conversion to open heart surgery.

^b Among 17 patients with unsuccessful device implantation, reasons included inability to deploy the valve (n = 4), adverse event (n = 2), improper device positioning (n = 2), device embolization (n = 1), and other (n = 8). Other reasons included suboptimal anatomic fit, operator decision to abort based on intraprocedural assessment, and catheter access challenges.

^c Of the 8 patients who underwent conversion to cardiac surgery, 4 had cardiac tamponade, 2 had ventricular rupture, 1 had device embolization, and the reason was unspecified for 1 patient.

^d Among 48 concomitant procedures, 7 were permanent pacemaker implantations, 1 was a peripheral intervention, 39 were procedure types not otherwise listed, and 1 was missing.

^e Eight patients required mechanical support (6 extracorporeal membrane oxygenation, 1 intra-aortic balloon pump, 1 unspecified).

Supplement 1. Subgroup comparisons used Pearson χ^2 tests, supplemented by generalized estimating equations, adjust-

ing for age and sex to address site-level clustering. Exploratory logistic regression models evaluated predictors of new CIED implantation among CIED-naïve patients and of bleeding events (eMethods in [Supplement 1](#)). Statistical significance used 2-sided tests, with significance set at $P < .05$. Analyses were performed with SAS version 9.4 (SAS Institute) at the Richard A. and Susan F. Smith Center for Outcomes Research at Beth Israel Deaconess Medical Center.

Results

From February 1, 2024, to March 31, 2025, TTVR procedures were attempted in 1034 patients at 82 sites. A valve was implanted in 1024 patients. Among the 10 patients in whom the valve was not implanted, 8 procedures were converted to surgery (eFigure 2 in [Supplement 1](#)). Results for the subset of patients with a valve implanted are presented in eTables 1 to 3 in [Supplement 1](#).

Baseline Characteristics

The mean (SD) age of the patients was 77.1 (10.6) years, 69.1% were female, 10.8% were Black, 83.2% were White, and 5.9% reported another race ([Table 1](#)). Race and ethnicity data were collected as part of the standardized STS/ACC TVT Registry forms; categories were self-reported by patients. At baseline, 73.2% of patients were in NYHA class III/IV, mean left ventricular ejection fraction was 55.9%, and 83.8% had atrial fibrillation or flutter. Diabetes was present in 29.7% of patients, liver disease in 12.3%, and prior CIED in 38.0%; 6.2% of patients were receiving dialysis and 48.4% had been hospitalized for heart failure in the prior year. The heart team classified the majority of patients as high (69.6% [686 of 985]) or extreme (10.5% [103 of 985]) surgical risk, with 15.1% (149 of 985) classified as intermediate risk and 4.8% (47 of 985) as low risk. TR was severe in 96.7% of patients, with the etiology reported as primary in 41.2%, secondary in 43.2%, CIED induced in 9.4%, and other in 6.1%. Moderate or severe mitral regurgitation was present in 39.9% of patients. Medications included diuretics in 83.9% of patients, anticoagulants in 68.1%, and antiplatelets in 36.1% ([Table 1](#)).

Procedural and Clinical Outcomes

Most procedures were performed using transfemoral venous access (97.4%) and under general anesthesia (98.5%). A valve was successfully implanted in 1017 (98.4%) of attempted cases. The median (IQR) procedure time was 103.0 (81.0-130.0) minutes. Valve size data were available in 1024 patients: 44 mm in 19.7%, 48 mm in 30.0%, 52 mm in 34.9%, and 56 mm (introduced later than the other sizes) in 15.4% ([Table 2](#)).

In-hospital all-cause mortality occurred in 2.3% of patients (24 of 1034), stroke in 0.2% (2 of 1034), and bleeding in 6.9% (71 of 1034). A new permanent CIED was implanted in 7.8% (81 of 1034) of all patients and 12.3% (79 of 641) of patients without a CIED at baseline. Device migration occurred in 0.6% (6 of 1034) of patients, embolization in 0.1% (1 of 1034), and 0.5% (5 of 1034) required tricuspid valve reintervention ([Table 3](#)). The majority of patients (93.8% [950 of 1013]) were

Table 3. In-Hospital and 30-Day Clinical Outcomes

Outcome	No. (%)	
	30-Day (n = 1034)	In-hospital (n = 1034)
All-cause mortality	32 (3.1)	24 (2.3)
Cardiovascular mortality ^a	21 (2.0)	17 (1.6)
Hemorrhagic stroke	1 (0.1)	1 (0.1)
Ischemic stroke	1 (0.1)	1 (0.1)
Heart failure hospitalization	32 (3.1)	NA
Adverse events		
New pacemaker or CIED		
In all patients	104 (10.1)	81 (7.8)
In patients without preexisting CIED, No./total No. (%)	102/641 (15.9)	79/641 (12.3)
Bleeding	82 (7.9)	71 (6.9)
Access site	14 (1.4)	14 (1.4)
Gastrointestinal	8 (0.8)	8 (0.8)
Genitourinary	3 (0.3)	3 (0.3)
Hematoma at access site	13 (1.3)	13 (1.3)
Retroperitoneal	8 (0.8)	8 (0.8)
Other ^b	30 (2.9)	30 (2.9)
Life-threatening or major bleeding ^c	13 (1.3)	NR
Vascular complications		
Minor	26 (2.5)	20 (1.9)
Major	14 (1.4)	11 (1.1)
Atrial fibrillation	25 (2.4)	20 (1.9)
Surgery/interventions		
Cardiac surgery or intervention - other unplanned	23 (2.2)	20 (1.9)
Vascular surgery or intervention - unplanned	16 (1.5)	12 (1.2)
Tricuspid valve reintervention ^d	5 (0.5)	5 (0.5)
Cardiac arrest	13 (1.3)	13 (1.3)
Dialysis (new requirement)	14 (1.4)	13 (1.3)
Pacemaker lead dislodgement/dysfunction	4 (0.4)	4 (0.4)
Cardiac perforation	6 (0.6)	6 (0.6)
Annular rupture	1 (0.1)	1 (0.1)
Coronary artery compression	1 (0.1)	1 (0.1)
Device-related events		
Embolization	1 (0.1)	1 (0.1)
Migration	6 (0.6)	6 (0.6)
Thrombosis	0	0
Myocardial infarction	2 (0.2)	0

Abbreviations: CIED, cardiac implantable electronic device; NA, not available; NR, not reported.

^a Determined from cause of death field on case report form. Deaths attributed to acute myocardial infarction, sudden cardiac death, heart failure, stroke, cardiovascular procedure, cardiovascular hemorrhage, other cardiovascular reason, or hemorrhage were classified as cardiovascular mortality; all other causes were classified as noncardiovascular mortality.

^b Bleeding from a site not specified (eg, pulmonary bleeding or subdural hematoma [not hemorrhagic stroke]) associated with any of the following from the medical record: (1) hemoglobin drop of ≥ 3 g/dL; (2) transfusion of whole blood or packed red blood cells; and (3) procedural intervention or surgery at bleeding site to reverse, stop, or correct bleeding (eg, surgical closures, exploration of arteriotomy site, or balloon angioplasty to seal arterial tear).

^c Only collected at follow-up.

^d Patient returned to operating room or cardiac catheterization laboratory for any tricuspid valve reintervention.

discharged home, and the median (IQR) postprocedure hospital stay was 2.0 (2.0-5.0) days. The total median (IQR) hospital length of stay, including prehabilitation, was 4.0 (2.0-6.0) days (Table 2). At 30 days, all-cause mortality occurred in 3.1% (32 of 1034) of patients, stroke in 0.2% (2 of 1034), bleeding in 7.9% (82 of 1034), and 3.1% (32 of 1034) of patients were hospitalized for heart failure (Table 3). New CIEDs were implanted in 10.1% (104 of 1034) of all patients and 15.9% (102 of 641) of patients without a CIED at baseline.

Echocardiographic and Hemodynamic Outcomes

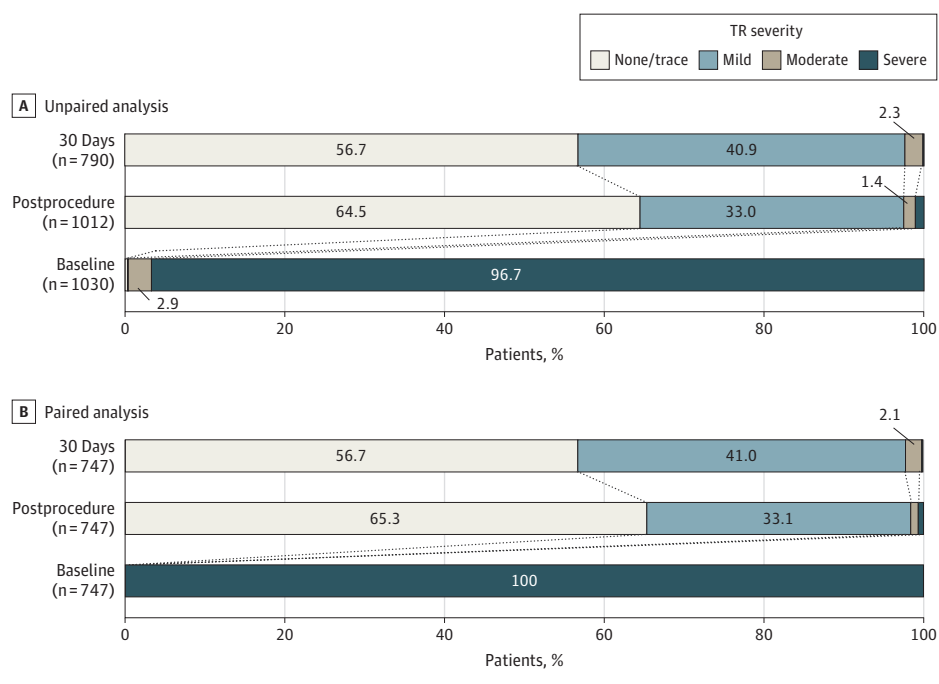
Among the 747 patients with paired baseline and 30-day echocardiography outcomes, 98.4% achieved mild or less TR, with 65.3% having none or trace TR ($P < .001$) post procedure (eTable 4 in Supplement 1). This significant improvement in

TR was sustained at 30 days, with 97.7% of patients with mild or less TR ($P < .001$). Unpaired analysis yielded similar results (Figure 1). In paired analysis from baseline to 30 days, mean (SD) right ventricle end-diastolic mid-diameter significantly decreased from 3.9 (0.9) cm to 3.6 (0.9) cm ($P = .003$) and right ventricle systolic pressure decreased from 41.1 (14.4) mm Hg to 33.8 (10.9) mm Hg ($P < .001$). Sensitivity analyses for paired comparisons using nonparametric tests are presented in eTable 5 in Supplement 1.

Functional and Quality-of-Life Outcomes

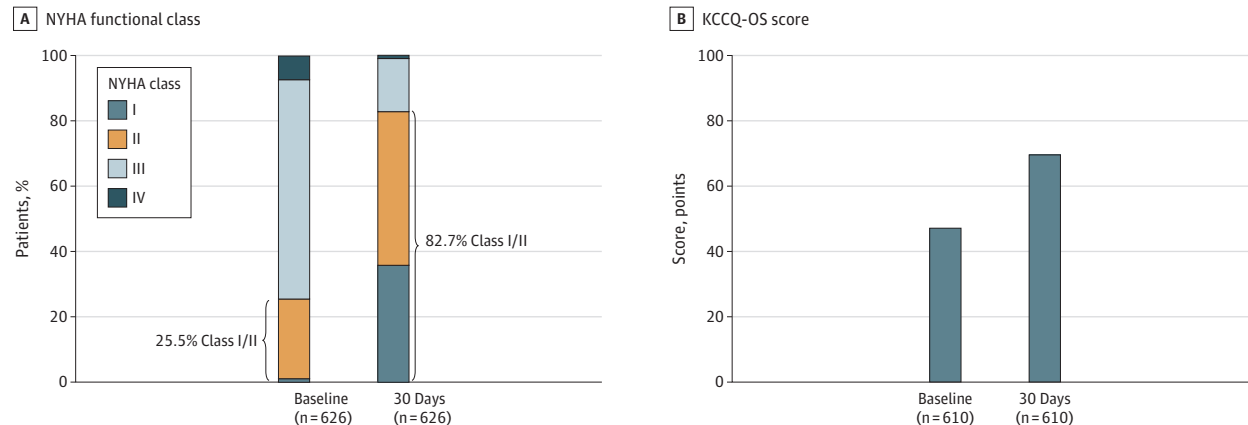
Patients experienced significant improvements in functional and quality-of-life outcomes in paired analyses between baseline and 30 days. At 30 days, of the 626 patients with paired data, 82.7% were in NYHA functional class I/II, compared with

Figure 1. Component Bar Graphs of Reduction in Tricuspid Regurgitation (TR) Severity at Discharge and 30 Days With Transcatheter Tricuspid Valve Replacement



Of the 1034 patients in the analysis population, 30-day, postprocedure, and baseline TR severity assessments were available for 790, 1012, and 1030 patients, respectively. Paired analysis included 747 patients who had TR severity assessments for all time points.

Figure 2. Component Bar Graphs of New York Heart Association (NYHA) Functional Class and Kansas City Cardiomyopathy Questionnaire Overall Summary (KCCQ-OS) Score at 30 Days



A, The P value for baseline and 30 days was $P < .001$. B, The mean (SD) KCCQ-OS scores were 47.1 (22.6) at baseline and 69.5 (24.4) at 30 days. The P value for baseline and 30 days was $P < .001$ and $\Delta = 22.4$.

25.6% at baseline ($P < .001$) (Figure 2A). In the 610 patients with paired data, the mean KCCQ-OS score improved by 22.4 points ($P < .001$) (Figure 2B).

Analysis Stratified by Baseline CIED Status

Comparing patients with and without prior CIEDs, the 38.0% of patients who had preexisting CIEDs were older (mean [SD] age, 78.5 [9.9] vs 76.2 [10.9] years; $P < .001$) and had higher rates of atrial fibrillation (87.5% vs 79.7%; $P = .001$) and cardiomyopathy (31.3% vs 17.5%; $P < .001$). There were no differ-

ences in procedural characteristics or 30-day TR severity (eTable 6 in Supplement 1).

At 30 days, there were no significant differences in all-cause mortality (CIED, 2.5% [10 of 393] vs no CIED, 3.4% [22 of 641]; $P = .47$) or heart failure hospitalization (CIED, 3.1% [12 of 393] vs no CIED, 3.1% [20 of 641]; $P > .99$), despite a higher access-site bleeding rate in patients with a preexisting CIED (2.8% [11 of 393] vs 0.5% [3 of 641]; $P = .003$). Between baseline and 30 days, both groups experienced similar improvements in NYHA functional class (CIED, 80.2% [199 of 248] vs

no CIED, 84.4% [319 of 378] achieving functional class I/II; $P = .55$) and mean (SD) KCCQ-OS score (CIED, 21.4 [26.0] vs no CIED, 23.0 [24.9] points; $P = .49$).

Other Subgroup Analyses

Effectiveness outcomes were consistent across all subgroups, including stratifications by sex, race, TR etiology, and dialysis (eTables 7-10 in [Supplement 1](#)). All subgroups benefited from postprocedural and 30-day TR reduction, with more than 95% achieving less than mild TR. Outcomes did not differ significantly by race or sex for any end point evaluated. By TR etiology, the rate of new CIED implantation differed significantly across groups (primary, 12.4%; secondary, 9.5%; pacemaker induced, 2.1%; other, 13.1%; $P = .009$). Patients receiving dialysis had significantly higher rates of all-cause hospitalization compared with patients not receiving dialysis (32.8% vs 11.5%; $P = .004$); no other outcomes differed significantly by dialysis status after clustering adjustment.

Predictors of New CIED Implantation and Bleeding

Exploratory multivariable logistic regression analyses were performed to identify potential predictors of new CIED implantation among CIED-naïve patients and of any bleeding event. For new CIED implantation, larger device diameter (52 mm or 56 mm vs 44 mm or 48 mm) was the only significant predictor on multivariable analysis (odds ratio, 1.76 [95% CI, 1.10-2.82]; $P = .02$). No candidate predictor reached significance for bleeding events. Full results are presented in eTables 11 and 12 in [Supplement 1](#).

Medical Management

In an unpaired analysis, diuretic use remained similar (83.9%-79.4% at discharge), while anticoagulant use increased from 68.1% to 94.2%, with 75.0% of patients taking direct oral anticoagulants. Antiplatelet use rose from 36.1% to 43.9%. At discharge, 39.6% of patients were taking both an anticoagulant and antiplatelet (eTable 13 in [Supplement 1](#)).

Data Completeness

Baseline data were largely complete; echocardiographic and health status outcomes had the most missing data (eTable 14 in [Supplement 1](#)). Inverse probability weighting sensitivity analyses yielded similar results to complete-case findings (eTable 15 in [Supplement 1](#)).

Discussion

In this initial post-US Food and Drug Administration approval cohort of 1034 consecutive TTVR procedures, 4 key findings are reported. First, the hemodynamic goal of near elimination of TR was achieved in routine clinical practice, with more than 97% of patients achieving mild or less TR at 30 days. Second, in this cohort of older patients with significant comorbidity burden, patients reported clinically meaningful improvements in symptoms and health status. Third, 30-day mortality was 3.1%, and early safety was acceptable, with lower new CIED and bleeding rates compared with randomized clinical trial experience.

The majority of the patients were discharged home. Fourth, these outcomes were broadly consistent across major clinical subgroups, including those stratified by presence of CIED at baseline, TR etiology, and dialysis. Taken together, the findings suggest that introduction of TTVR after regulatory approval in the US has been safe and effective.

The TRISCEND II pivotal trial, which randomized 267 patients to undergo TTVR and 133 to undergo medical therapy, demonstrated that patients who were randomized to TTVR achieved durable near elimination of TR at 1-year follow-up, with more than 95% of patients achieving mild or less TR, along with significant clinical benefits driven by quality-of-life improvements.¹² Study analysis of the available data on TTVR from the STS/ACC TVT Registry provides a large dataset from routine US practice and shows similar TR elimination at an early stage: 97.7% of patients had mild or less TR at 30 days. Moreover, 82.7% of patients achieved NYHA functional class I/II, and mean KCCQ-OS scores improved by 22.4 points. The magnitude of TR elimination and improvements in patient-reported and echocardiographic outcomes align with those reported in TRISCEND II, supporting generalizability of the treatment effect outside of clinical trial settings.^{12,15,16}

A particular focus of safety in TTVR relates to new conduction abnormalities. New conduction disturbances requiring permanent pacing are a known risk for surgical tricuspid valve procedures, with reported rates in the range of 6% to 14% for repair and 15% to 34% for replacement.^{7,17-19} In the present study, 10.1% of all patients (15.9% of CIED-naïve patients) required a new CIED within 30 days, a comparably lower rate than in the TRISCEND II trial (15.4% and 24.7%, respectively)¹² and within the range of reports on self-expanding transcatheter aortic valve replacement of 10.8% to 18.3%.^{20,21} Exploratory multivariable analysis identified larger device diameter (52 mm or 56 mm vs 44 mm or 48 mm) as the only independent predictor of new CIED implantation (adjusted odds ratio, 1.76; $P = .02$). This finding is consistent with the hypothesis that a larger prosthesis may exert greater mechanical force on the conduction system adjacent to the tricuspid annulus.²²

Beyond the risk of new CIED implantation, the presence of baseline CIEDs introduced unique technical considerations during TTVR, including lead and leaflet interaction, impaired visualization, and the potential for lead entrapment.^{12,13} In this cohort, one-third of patients had a CIED at baseline; however, this did not translate into reduced procedural safety and effectiveness. Procedural duration and TR reduction were comparable regardless of CIED presence, bleeding and mortality rates remained low, and health status improved substantially across both groups, reinforcing the applicability of TTVR across a broad patient population.

The lower rate of bleeding (7.9%) at 30 days compared with TRISCEND II (10.4%) is encouraging and may be influenced by differences in baseline patient characteristics, anti-thrombotic use, maturation of imaging-guided implantation techniques, and evolution of periprocedural management. In TRISCEND II, postprocedural anticoagulation plus antiplatelet therapy was recommended in all patients up to 6 months, compared with approximately 40% of patients discharged with combined anticoagulation and antiplatelet therapy in the

STS/ACC TVT Registry, likely contributing to the lower severe bleeding rates reported in this study. The comparatively low bleeding rate in this real-world report is especially notable given that the bleeding definition is more inclusive than the Mitral Valve Academic Research Consortium severe bleeding definition used in TRISCEND II.¹² Reassuringly, the findings are directionally concordant with the European real-world analysis (176 patients, 12 centers), where TTVR achieved mild or less TR in 98% of patients at 30 days, clinically meaningful improvements in functional status, a CIED implantation rate of 18.9% among CIED-naïve patients, and low rates of life-threatening bleeding (1.7%) and major vascular complications (1.1%).²³

These data add to the growing evidence in the field of transcatheter tricuspid repair and replacement. Real-world T-TEER registries report modest bleeding (approximately 5%-8%), minimal need for new CIED (<1%), and 30-day mortality of approximately 1%; however, TR reduction to mild or less is achieved in only approximately 50% to 60% of patients.^{10,11} By contrast, in this predominantly high-surgical risk TTVR cohort, TR reduction was near universal: mild or less in approximately 98% of patients at 30 days, with larger health status gains, balanced against higher pacemaker rates among device-naïve patients (approximately 15%) and bleeding (7.9%). Utilization of TTVR may have a potential advantage over T-TEER in patients with large coaptation gaps, significant leaflet tethering, or preexisting transvenous leads. These complementary profiles support a pragmatic selection framework: consider T-TEER when residual TR is acceptable and pacing risk should be minimized, and consider TTVR when the therapeutic goal is near elimination of TR or anatomy is unfavorable for repair.²² Randomized head-to-head comparisons in well-defined patient subgroups will ultimately be required to establish evidence-based selection criteria for heart teams.

Limitations

This study has limitations. First, this observational, site-reported registry is subject to confounding, missing data, and potential inaccuracies. Paired echocardiographic TR analyses were available in approximately 72% of patients and paired KCCQ-OS data in approximately 59%; inverse probability weighting sensitivity analyses confirmed robustness of findings. Second, echocardiographic outcomes were site interpreted without independent core laboratory adjudication. Third, the STS/ACC TVT Registry does not capture all variables needed to calculate established surgical risk scores (STS score, European System for Cardiac Operative Risk Evaluation II, TRI-SCORE); surgical risk was instead characterized by the heart team's clinical risk classification. Fourth, CIED variables were derived from registry fields with limited granularity. Fifth, these findings reflect early US experience at 30 days and do not capture longer-term outcomes. Continued follow-up through the STS/ACC TVT Registry will be important to corroborate these early real-world results.

Conclusions

In this real-world study among an older population with significant comorbidities, consecutive patients with severe TR treated with TTVR achieved near elimination of TR, with substantial short-term gains in symptoms and health status and acceptable safety, including the need for CIED implantation and bleeding. Importantly, rates of new CIED implantation and severe bleeding were lower than the randomized clinical trial experience. Results were consistent across patients with and without CIEDs and other subgroups, supporting the broad applicability of TTVR in contemporary clinical practice.

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